Dear Editors,

I am submitting "Waveform Asymmetry as a Biomarker of Neural Aging: Spatial Degradation of Oscillatory Cycle Shape Across Two Independent Cohorts" for consideration as a Research Article in Network Neuroscience.

This manuscript addresses a gap in the EEG aging literature: while spectral power and frequency have been extensively studied, the *shape* of individual oscillatory cycles has been largely overlooked. We quantify waveform shape using peak-trough asymmetry (PTA) and report three principal findings:

1. **Band-specific spatial structure.** Slow and fast rhythms show opposing waveform asymmetry, with a robust spatial double dissociation between posterior alpha and central beta — reflecting distinct excitatory versus inhibitory generating circuits.

2. **Replicated aging biomarker.** Beta-band asymmetry declines with age (Cohen's $d = 0.7$), replicated across two independent cohorts (LEMON N=215; Dortmund N=608) with pre-registered blind predictions that were precisely confirmed. The effect survives control for aperiodic spectral slope and EMG artifacts. Longitudinal 5-year follow-up (N=208) confirms test-retest reliability.

3. **Spatial network degradation.** The spatial organization of waveform asymmetry simplifies with age, and theta spatial entropy independently predicts memory performance — linking oscillatory cycle shape to functional cognitive networks.

The work fits Network Neuroscience because it examines spatial network-level organization of a novel neural metric, demonstrates cross-system coupling between brain and cardiac oscillatory shape, and provides all data and code openly (https://github.com/mool32/waveform-asymmetry-aging). A preprint has been posted on bioRxiv.

Suggested reviewers:

- Bradley Voytek (University of California, San Diego) — originator of the PTA method; expertise in waveform shape and neural oscillations
- Natalie Schaworonkow (University of Tuebingen) — expertise in waveform shape across development and spatial synchronization
- Thomas Donoghue (Columbia University) — expertise in parameterizing neural power spectra and aperiodic activity

This manuscript is original and not under consideration at another journal. The author has approved the submission.

Sincerely, Theodor Spiro Vaika Inc., East Aurora, NY, USA theospirin@gmail.com